

EXPERIMENT 13:

AIM:

To generate a parse tree for a given sentence using a context-free grammar.

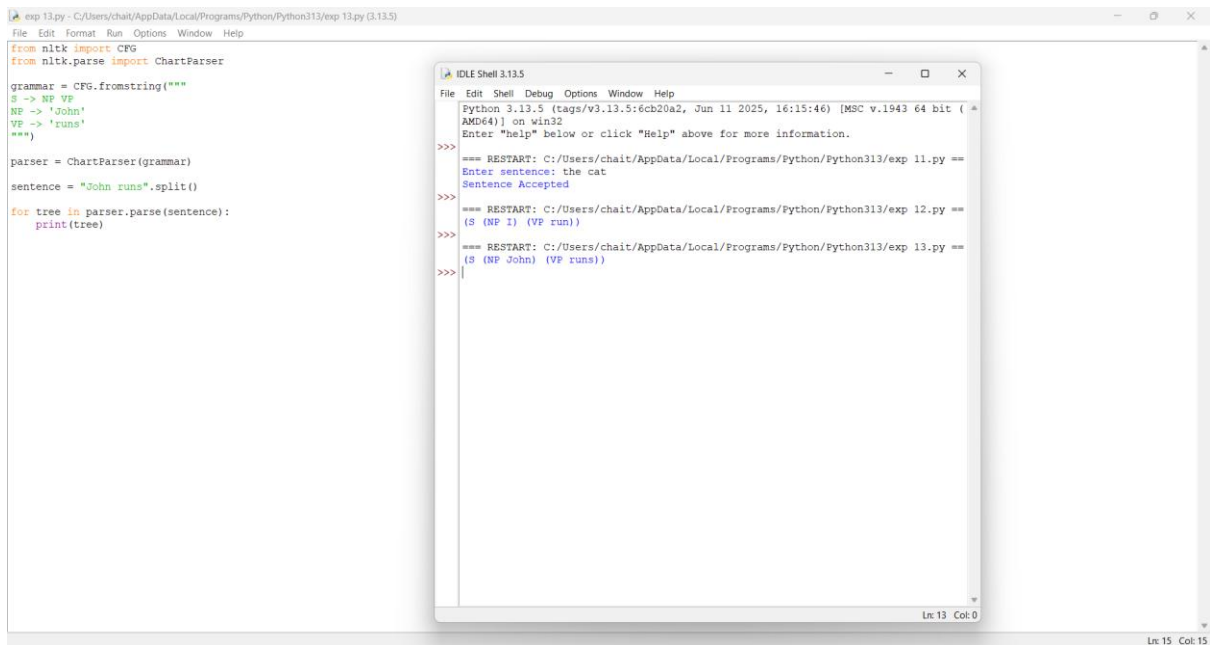
PROGRAM:

```
from nltk import CFG
from nltk.parse import ChartParser

grammar = CFG.fromstring("""
S -> NP VP
NP -> 'John'
VP -> 'runs'
""")

parser = ChartParser(grammar)
sentence = "John runs".split()
for tree in parser.parse(sentence):
    print(tree)
```

OUTPUT:



The screenshot displays a Python IDE window titled 'exp 13.py - C:/Users/chaht/AppData/Local/Programs/Python/Python313/exp 13.py (3.13.5)'. The code in the editor is as follows:

```
from nltk import CFG
from nltk.parse import ChartParser

grammar = CFG.fromstring("""
S -> NP VP
NP -> 'John'
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""")

parser = ChartParser(grammar)
sentence = "John runs".split()
for tree in parser.parse(sentence):
    print(tree)
```

Overlaid on the IDE is an 'IDLE Shell 3.13.5' window showing the execution output:

```
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
=== RESTART: C:/Users/chaht/AppData/Local/Programs/Python/Python313/exp 11.py ===
Enter sentence: the cat
Sentence Accepted
>>>
=== RESTART: C:/Users/chaht/AppData/Local/Programs/Python/Python313/exp 12.py ===
(S (NP I) (VP run))
>>>
=== RESTART: C:/Users/chaht/AppData/Local/Programs/Python/Python313/exp 13.py ===
(S (NP John) (VP runs))
>>>
```

The status bar at the bottom right of the IDE shows 'Ln: 15 Col: 15'.